

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%

Assessment Tool Weightage: 25%

QUESTIONS:

Total Marks: 25

Case Study 1: CNN for Automated Medical Image Classification

A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.
A) A suitable CNN architecture for chest X-ray classification can begin with an input layer that accepts resized X-ray images, such as 224×224 pixels. The image is passed through several convolutional layers with ReLU activation to extract features such as edges, textures, shapes and abnormal regions. Pooling layers are then used to reduce the spatial dimensions and computational complexity. After several convolution and pooling stages, the extracted feature maps are converted into a compact feature representation using Global Average Pooling or Flattening. Finally, fully connected layers perform classification, and a sigmoid output layer can be used to classify the image into Normal or Disease-Affected categories.

2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.

A) CNNs are suitable for chest X-ray classification because they can automatically learn important visual features without manually designing them. **Convolutional layers** apply small filters to the X-ray image and detect features such as edges, textures and abnormal patterns. **Filters** are learnable matrices that scan different parts of the image and generate feature maps. **Pooling layers**, such as max pooling, reduce the size of feature maps while retaining important information, which reduces computation and helps make the model less sensitive to small changes in position. **Fully connected layers** combine the learned features and use them to make the final classification. The ReLU activation function adds non-linearity, allowing the CNN to learn complex patterns present in medical images.

3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.

A) Parameter sharing is an important feature of CNNs that significantly reduces the number of trainable parameters. Instead of using different weights for every pixel location, a CNN uses the same filter weights across different regions of the image. For example, a 3×3 filter contains only 9 weights plus a bias, and this same filter can scan the entire X-ray image to detect a particular feature. In contrast, a fully connected network would require separate connections between almost every input pixel and neuron. Therefore, parameter sharing reduces memory and computational requirements, makes training faster, and helps reduce overfitting while allowing the CNN to detect the same feature wherever it occurs in the X-ray.

4. Discuss suitable regularization techniques such as dropout, data augmentation and batch normalization to reduce overfitting.

A) Regularization is important because a deep CNN may overfit the training X-ray images and perform poorly on new patients. Dropout randomly deactivates some neurons during training, forcing the network to learn more robust features instead of depending on specific neurons. Data augmentation increases the variety of training images by applying suitable transformations such as small rotations, shifts, scaling and brightness adjustments. For medical images, these transformations should be clinically realistic. **Batch normalization** normalizes the activations during training, making the learning process more stable and helping the model converge faster. Other techniques such as early stopping and L2 weight regularization can also be used to improve generalization.

5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

A) Between AlexNet and ResNet, **ResNet would generally be more appropriate** for chest X-ray classification. AlexNet is an earlier CNN architecture with fewer layers and can require fewer computational resources, but its feature-learning capability is more limited compared with modern deep networks. ResNet uses a deeper architecture with **residual or skip connections**, which allow information and gradients to pass through the network more effectively. This enables ResNet to learn complex hierarchical features, which is useful for identifying subtle abnormalities in chest X-rays. Although ResNet requires more computational resources, pretrained models such as **ResNet-18 or ResNet-50** can use transfer

learning to reduce training requirements. Therefore, ResNet is preferred when better feature learning and classification performance are important.

Case Study 2: CNN-Based Autonomous Vehicle Object Recognition

An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.

A) For an autonomous vehicle vision system, a CNN can be designed to process camera images and identify multiple objects such as **cars, pedestrians, bicycles, traffic signs, and road obstacles**. The input images can first be resized and normalized, after which they are passed through several convolutional layers to extract visual features. Early layers detect simple patterns such as edges and corners, while deeper layers identify complex patterns such as wheels, human shapes, vehicles and traffic signs. Pooling or suitable downsampling layers reduce the spatial size of feature maps and computational cost. Finally, fully connected or detection-specific output layers can classify the detected objects. For a real autonomous vehicle, an object-detection model rather than simple image classification would normally be required because the system must identify both **what the object is and where it is located** in the image.

2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.

A) The **input layer** receives camera images and converts them into a suitable numerical format after resizing and normalization. **Convolutional layers** apply learnable filters to different regions of the image to extract useful visual patterns. **Activation functions**, such as ReLU, introduce non-linearity and allow the network to learn complex relationships. **Pooling layers** reduce the spatial dimensions of feature maps while retaining important information, thereby reducing computational requirements. The **fully connected/output layers** use the extracted features to produce the final object classes or detection results. CNN filters progressively learn features: the first layers generally learn **low-level features** such as edges, lines and corners; middle layers combine these to learn textures, shapes and parts; and deeper layers learn **high-level features** such as complete cars, pedestrians, bicycles and traffic signs.

3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.

A) Parameter sharing means that the same convolutional filter is used across different locations of an image instead of learning separate weights for every location. This greatly reduces the number of trainable parameters and therefore lowers memory usage and computational complexity. For example, a small 3×3 filter can scan the complete camera image using the

same weights. If the filter learns to recognize an edge or another useful pattern, it can detect that pattern wherever it appears in the image. This property provides translation-related feature detection, meaning that an object feature can be recognized even when it changes its position within the image. Parameter sharing is especially useful in autonomous vehicles because objects can appear at different positions, distances and viewing angles.

4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.

A) Overfitting occurs when the CNN learns the training images too closely, including irrelevant details or noise, and therefore performs poorly on new road scenes. This can be a problem because autonomous vehicles encounter many different environments, weather conditions, lighting levels and object appearances. **Data augmentation** can be used by applying realistic changes such as cropping, scaling, brightness adjustment, small rotations and horizontal flipping where appropriate. **Dropout** can randomly deactivate neurons during training and prevent excessive dependence on particular features. **Batch normalization** helps stabilize training and can provide some regularization. **L2 regularization or weight decay** can also prevent excessively large weights, while **early stopping** can stop training when validation performance stops improving. These techniques help the model generalize better to unseen road conditions.

5. Compare **AlexNet** and **ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.

A) **AlexNet** is an earlier CNN architecture with a relatively shallow network compared with modern architectures. It uses convolutional and pooling layers followed by fully connected layers and has a comparatively simpler structure. **ResNet**, on the other hand, can contain many more layers and uses **residual or skip connections** to make deep networks easier to train. AlexNet has a simpler architecture and generally lower computational requirements, but its feature-learning capability is more limited. ResNet can learn more complex and hierarchical features and generally provides better accuracy on difficult visual recognition tasks. ResNet's skip connections also reduce the **vanishing-gradient problem**, which can make training deep networks easier. Although ResNet may contain many layers, modern ResNet variants can be more efficient than older architectures because of improved design and the ability to reuse pretrained models.

6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.

A) For an autonomous vehicle system, ResNet would generally be the better choice because it provides stronger feature-learning capability and higher recognition accuracy for complex objects and environments. Its residual connections allow deeper networks to learn effectively and reduce problems associated with vanishing gradients. However, real-time performance is also critical because the vehicle must process camera frames quickly. Therefore, instead of using a very large ResNet model, a smaller variant such as ResNet-18 can be considered when computational resources are limited. A pretrained ResNet backbone can also be combined with an appropriate object-detection architecture to identify both the class and location of objects. Thus, ResNet offers a better balance of accuracy and feature learning, while the exact model

size should be selected according to the vehicle's hardware and real-time processing requirements.

Code of Conduct:

I K. Hema sai Siddartha(192325036) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Hema sai Siddartha

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand